

Lecture 5 - 6: Circuit Analysis - KVL, Loop Analysis

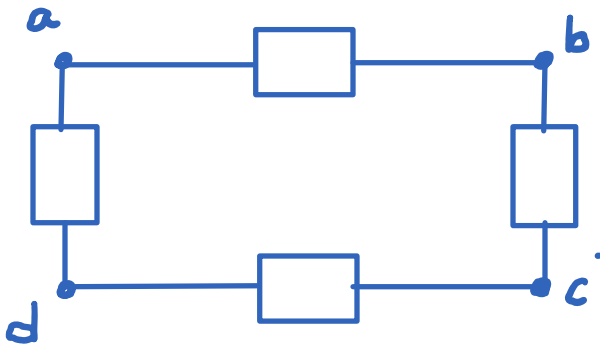
Learning Objectives:

1. Define Kirchhoff's voltage law (KVL)
2. Compute currents in simple circuits using KVL and Ohm's law
3. Use loop analysis method to compute loop currents
4. Derive voltage division formula and analyze the limitations of of voltage divider

In practice we can encounter circuits that may have several electrical elements. Ohm's law defines the relationship between current and voltage in a resistor. Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL) help us in defining relationships between current and voltages in more complex circuits. These relationships are useful in solving for unknown voltages and currents in a circuit.

1. Kirchhoff's Voltage Law(KVL)

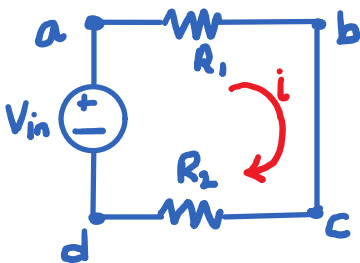
KVL states that the algebraic sum of voltages around a loop is zero. As we discussed earlier, Loop is a closed path in a circuit starting at a node, traversing through a series of nodes, and ending at the starting node without passing through the same node twice. Consider the circuit with one loop shown below. The sum of voltages while passing through the clockwise direction must sum to zero as shown.



$$\text{KVL: } V_{ab} + V_{bc} + V_{cd} + V_{da} = 0$$

Fig 5.1. Illustration of KVL in a loop.

Example 1: Write the KVL equation for the circuit shown below.



$$\text{KVL: } V_{ab} + V_{bc} + V_{cd} + V_{da} = 0$$

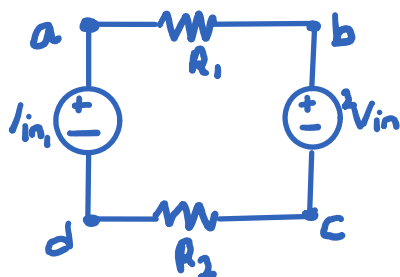
$$\text{Ohm's Law: } i_{ab}R_1 + 0 + i_{cd}R_2 - V_{in} = 0$$

$$\Rightarrow iR_1 + iR_2 = V_{in}$$

$$\Rightarrow \frac{V_{in}}{i} = R_1 + R_2.$$

What does the quantity on the right hand side of the above equation represent?

Example 2:



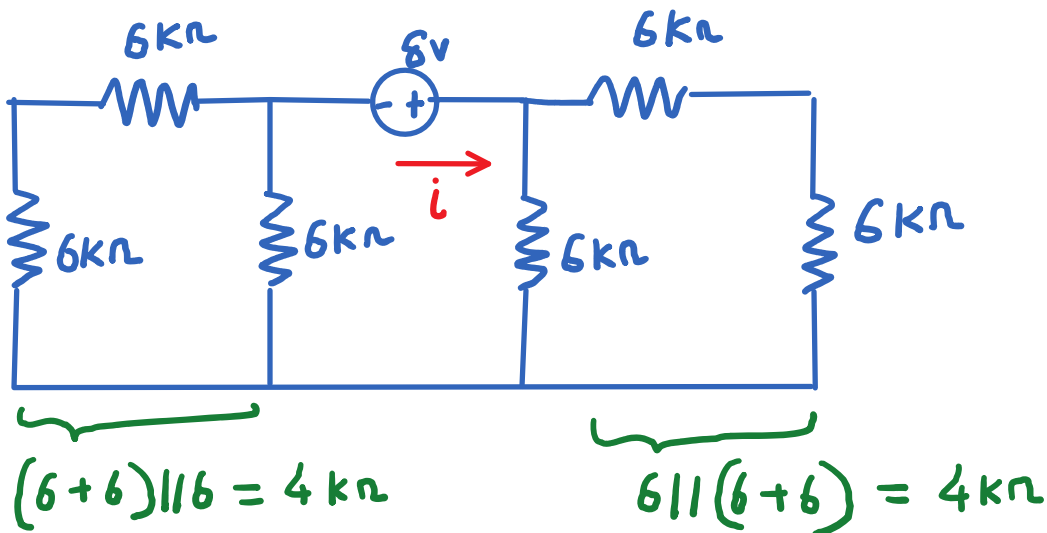
$$\text{KVL: } V_{ab} + V_{bc} + V_{cd} + V_{da} = 0$$

$$\text{Ohm's Law: } i_{ab}R_1 + V_{in2} + i_{cd}R_2 - V_{in1} = 0$$

$$\Rightarrow iR_1 + iR_2 = V_{in1} - V_{in2}$$

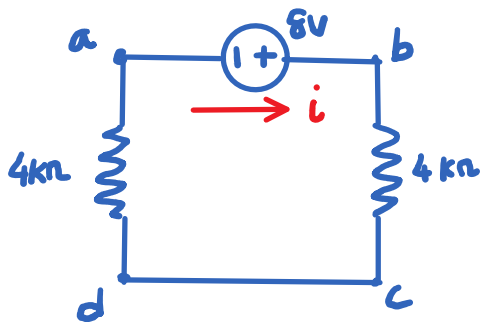
$$\Rightarrow \frac{V_{in1} - V_{in2}}{i} = R_1 + R_2.$$

Example 3: Find current i in the circuit shown below.



Solution:

The circuit can be redrawn as shown below:



KVL: $V_{ab} + V_{bc} + V_{cd} + V_{da} = 0$

Ohm's Law: $-8 + i_{bc} \times 4 + 0 + i_{da} \times 4 = 0$

$\Rightarrow 4i + 4i = 8$

$\Rightarrow i = 1\text{mA.}$

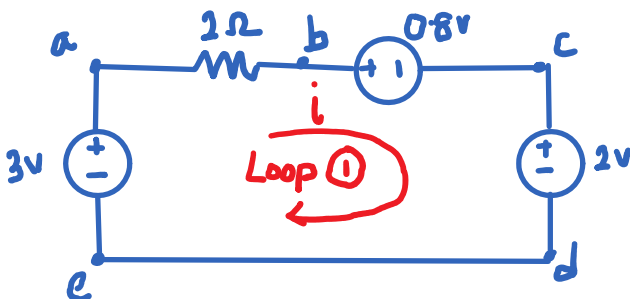
2. Loop analysis

We looked at Kirchhoff's voltage law and applied it for simple circuits containing one loop. Loop analysis is a systematic procedure based on KVL to solve for currents in more complex circuits. Loop current analysis involves the following steps:

- Identify loops in a circuit.
- Pick currents in clockwise direction.
- Set up loop equations.
- Solve system of equations to obtain unknown currents.

Example 2.1: Consider the single loop circuit shown below. Obtain the unknown current i .

Solution:



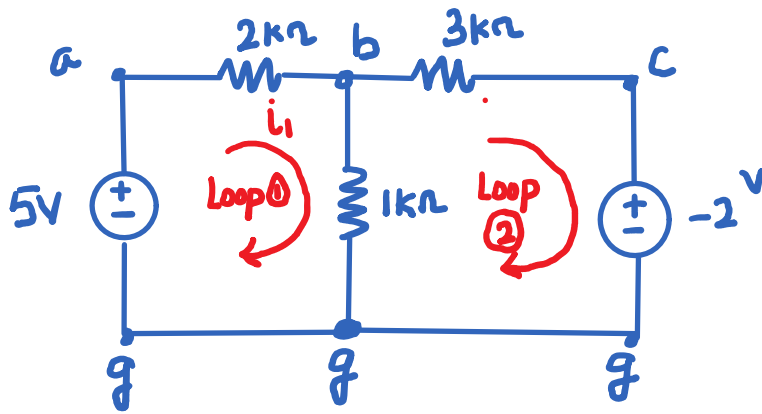
Solution:

KVL: $V_{ab} + V_{bc} + V_{cd} + V_{ea} = 0$

Ohm's Law: $2i + 0.8 + 2 - 3 = 0$

$\Rightarrow i = 0.1\text{ A}$

Example 2.2: Compute the currents i_1, i_2 in the circuit shown below.



Solution:

Loop 1(KVL): $V_{ab} + V_{bg} + V_{ga} = 0$

Ohm's Law: $2i_{ab} + 1i_{bg} - 5 = 0$

$$\Rightarrow 2i_1 + 1(i_1 - i_2) - 5 = 0$$

$$\Rightarrow 3i_1 - i_2 = 5 \quad (1)$$

Loop 2(KVL): $V_{bc} + V_{cg} + V_{gb} = 0$

Ohm's Law: $3i_{bc} - 2 + 1i_{gb} = 0$

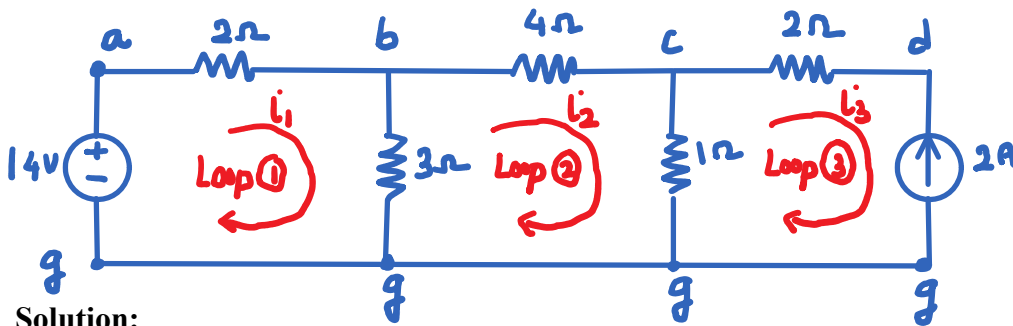
$$\Rightarrow 3i_2 + 1(i_2 - i_1) - 2 = 0$$

$$\Rightarrow -i_1 + 4i_2 = 2 \quad (2)$$

Solve (1) and (2): $\begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ mA.}$$

Example 2.3: Compute currents i_1, i_2, i_3 in the circuit shown below.



Solution:

By observation: $i_3 = -2 \text{ A}$

Note: We cannot write a loop equation for loop 3. Why

Loop 1(KVL): $V_{ab} + V_{bg} + V_{ga} = 0$

Ohm's Law: $2i_{ab} + 3i_{bg} - 14 = 0$
 $\Rightarrow 2i_1 + 1(i_1 - i_2) - 14 = 0$
 $\Rightarrow 3i_1 - i_2 = 14 \quad (1)$

Loop 2 (KVL): $V_{bc} + V_{cg} + V_{gb} = 0$

Ohm's Law: $4i_{ab} + 1i_{bg} + 3i_{gb} = 0$
 $\Rightarrow 4i_2 + 1(i_2 - i_3) + 3(i_2 - i_1) = 0$
 $\Rightarrow -3i_1 + 8i_2 - i_3 = 0$
 $\Rightarrow -3i_1 + 8i_2 = -2 \quad (2)$

Solve (1) and (2): $\begin{bmatrix} 3 & -1 \\ -3 & 8 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 14 \\ -2 \end{bmatrix} \Rightarrow \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -3 & 8 \end{bmatrix}^{-1} \begin{bmatrix} 14 \\ -2 \end{bmatrix} = \begin{bmatrix} 5.42 \\ 2.28 \end{bmatrix} \text{ A}$